

Homework 4: Imitation Learning

CS/Stat 184(0): Introduction to Reinforcement Learning

0 Instructions

For each question in this HW, please list all your collaborators and reference materials (beyond those specified on the website) that were used for this homework. Please add your remarks in a “Question 0”.

1 Off-policy Policy Gradient Estimation [15 points]

In this section, we are going to study the off-policy PG formulation. We consider the following Finite horizon Markov Decision Process $\mathcal{M} = \{\mathcal{S}, \mathcal{A}, H, r, P, \mu_0\}$ where μ_0 is the initial state distribution, i.e., $s_0 \sim \mu_0$. We are interested in computing the policy gradient of a policy $\pi : \mathcal{S} \mapsto \Delta(\mathcal{A})$ (i.e., π is a stochastic policy mapping from state to distribution over actions, $a \sim \pi(\cdot|s)$).

1.1 Background and Setting

In the lecture, we talked about how to estimate the policy gradient of π using samples/trajectories sampled from π itself, which is called on-policy estimation, i.e., use trajectories/samples from a different policy $\pi^b : \mathcal{S} \mapsto \Delta(\mathcal{A})$ to estimate the policy gradient of π . The off-policy setting is useful, for example, when we have a large amount pre-collected offline data and we do not want to execute π in the real system due to some safety concern (think about medical application: we may have large logged data from previous patients, and we do not want to directly collect new samples by deploying some policy π , e.g., a new treatment, on new patients due to safety reasons). We instead hope to estimate the outcome of the new policy using the offline data collected under a different policy.

Denote $\tau^1, \dots, \tau^N$ as N trajectories generated by the behavior policy π^b , i.e.,

$$\tau^i = \{s_h^i, a_h^i, \pi^b(a_h^i|s_h^i), r_h^i\}_{h=0}^{H-1},$$

where $s_0^i \sim \mu_0$, $a_h^i \sim \pi^b(\cdot|s_h^i)$, $s_{h+1}^i \sim P(\cdot|s_h^i, a_h^i)$, $r_h^i = r(s_h^i, a_h^i)$. These N trajectories form our offline dataset. We are going to assume that π^b is diverse enough to have sufficient support on π , i.e.,

$$\forall s, a, \frac{\pi(a|s)}{\pi^b(a|s)} < C, \text{ where } C \in \mathbb{R}^+.$$

1.2 Unbiased Estimator of Off-policy PG

Below we show that how we can use the offline dataset to compute an unbiased estimate of the gradient of π , without collecting any samples/trajectories from π itself. For notation simplicity, we drop the parameterization on π , and simply denote $\nabla \ln \pi(a|s)$ as the gradient of the log-likelihood of π with respect to its internal parameters (i.e., we use $\nabla \ln \pi(a|s)$ in short of $\nabla_{\theta} \ln \pi_{\theta}(a|s)$).

Q1 [15 points] Prove that the following estimator is an unbiased estimate of the original PG of π :

$$\frac{1}{N} \sum_{i=1}^N \left(\prod_{h=0}^{H-1} \frac{\pi(a_h^i | s_h^i)}{\pi^b(a_h^i | s_h^i)} \right) \sum_{h=0}^{H-1} \nabla \ln \pi(a_h^i | s_h^i) (R(\tau^i)), \quad (1)$$

where $R(\tau^i) = \sum_{h=0}^{H-1} r_h^i$, i.e., the total reward of trajectory τ^i (In other words, you just need to derive the formulation with the right expectation corresponding to off-policy).

With the above unbiased estimator, we can simply run stochastic gradient ascent algorithm to optimize policy in a completely offline manner, without ever interacting with the system to get online samples. One might also want to apply variance reduction tricks in practice.